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Population balance modelling of protein precipitates exhibiting turbulent flow through a circular pipe

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ABSTRACT

The breakage of liquid–liquid, solid–liquid and solid–gas dispersions occurs in many industrial processes during the transport of particulate materials. In this work, breakage of whey protein precipitates passing through a capillary pipe is examined and an experimentally derived breakage frequency is applied to construct a suitable population balance model to characterize the breakage process. It has been shown that the breakage frequency of precipitate particles is highly dependent on their shear history and on the turbulent energy dissipation rate in the pipe. The population balance equation (PBE) uses a volume density based discrete method which is adapted from mass density based discretization. In addition to comparing the model with experimental data, predicted results at different velocities are presented. It was found that the population balance breakage model provides satisfactory results in terms of predicting particle size distributions for such processes.

1. Introduction

Population balance modelling (PBM) is a useful predictive tool which can be used to analyze particle size distribution (PSD) over time for processes in the food, chemical and pharmaceutical industries. Population balances trace their origins from the Boltzmann (1872) equation which was developed in 1872. However, the application of population balances to process engineering systems is a relatively new development. Early examples include Hulburt and Katz (1964) who applied their model to dispersed phase systems, while Randolph and Larson (1964) primarily focused on crystallization. Since then, there have been many articles published dealing with population balances as applied to process engineering systems.

Population balance equations (PBEs) are underpinned by the law of conservation of mass and the equations describe the relationships that define the number balance on particles of given particulate states. Alternatively, the number balance can also be considered in terms of mass or volume for some instances. The structure of the population balance equation entails partial integro-differential equations. The framework of these equations is complex and analytical solutions may be impossible. However, some simplified approaches which allow for population balance equations to be solved are available in the literature (Gelbard and Seinfeld, 1978; Bapat et al., 1983; Hounslow et al., 1988; Hill and Ng, 1995; Kumar and Ramkrishna, 1996; Nicmanis and Hounslow, 1998; Vanni, 1999; Kostoglou and Karabelas, 2002; Diemer and Olson, 2002; Marchisio et al., 2003). These approaches help simplify the rigorous population balance equations and include mathematical methods such as Monte-Carlo simulation, method of moments, discrete formulations and Laplace transforms. More detailed reviews and solution techniques of PBEs are available in Ramkrishna (2000).

PBEs are used to define phenomena such as nucleation, growth, aggregation and breakup of particles. This work is concerned with aggregate breakage only. The term breakage (fragmentation) can be defined as separation of fragments from a whole particle. Breakage occurs in many engineering applications, such as mixing, conveying operations, liquid–liquid dispersion, milling and grinding applications, crystal and precipitate production, transport and separation. Modelling a breakage process can be achieved by constructing a PBE in the form of breakage functions.

There has been much research on breakage of dispersions (Coulaloglou and Tavlarides, 1977; Karabelas, 1978; Narsimhan et al., 1979; Ayazi Shamlou et al., 1994; Peng and Williams, 1994; Luo and Svendsen, 1996; Wojcik and Jones, 1998; Kramer and Clark, 1999; Blaser, 2000; Byrne and Fitzpatrick, 2002; Nere and Ramkrishna, 2005; Kostoglou, 2006; Kostoglou and Karabelas, 2007). Although there are numerous investigations that have been carried out on breakage in turbulent flow, a few of these studies deal with the breakage of flowing dispersions. More recent publications

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Nere and Ramkrishna (2005), Kostoglou (2006) and Kostoglou and Karabelas (2007) studied the breakage in turbulent pipe flow. However, these recent studies dealt with breakage of liquid–liquid dispersions. On the other hand, there are a number of examples of breakage study in liquid–solid dispersions including Ayazi Shamlou et al. (1994), Kramer and Clark (1999), Blaser (2000), Wojcik and Jones (1998), Byrne and Fitzpatrick (2002) and Zumaeta et al. (2006). Moreover, single particle breakage of aggregates has been widely investigated by many researches (Yuregir et al., 1987; Thorntony et al., 1996; Subero and Ghadiri, 2001; Salman et al., 2003). Additionally, a recent book edited by Salman et al. (2007) provides a brief overview of particle breakage from the small scale of a single particle, to the study of whole processes for breakage; both by experimental study and mathematical modelling.

A study of whey protein precipitate breakage was carried out by Zumaeta et al. (2006). They developed a breakage model for predicting the mean size of the particles due to breakage as a result of turbulent flow through various pipe geometries. Computational fluid dynamic (CFD) simulations were applied to evaluate the turbulent eddy dissipation rate and corresponding exposure time.

In this work, a population balance is applied to the experimental data associated with particle breakage as a result of a precipitate dispersion flowing through the high-shear capillary rig as described in Zumaeta et al. (2006). Population balance equations are constructed to describe the breakage of whey protein precipitates for volume fraction density using a discretization method. The PBEs that govern this breakage process are solved using a volume density based discrete method which is adapted from mass density based discretization. In addition to comparing the experimental results with the model, predicted results at different velocities are also presented.

2. Population balances for breakage

Modelling a breakage process can be achieved by constructing a population balance equation in the form of breakage functions. This can be either in continuous or in discrete form. Moreover, two types of breakage equations are defined:

(i) a number density based breakage equation,
(ii) a mass density based breakage equation.

When particles display homogenous density then a mass density based breakage equation can be considered as a volume density based breakage equation. In this study, it is assumed that particles are spherical and of broadly homogenous density, thus a mass density based breakage equation will be used to model volume fraction.

2.1. Breakage equation

The breakage of liquid–liquid, solid–liquid and solid–gas dispersions occurs in many industrial processes during the transport of particulate materials. The transportation of particulate materials is usually accomplished by means of cylindrical pipes. A general form of the equation that represents the evolution of particle breakage in the cylindrical pipes can be defined as

$$u(r)\frac{\partial\hat{\phi}(x,r,z)}{\partial z} = \int_x^\infty p(x,y,r)b(y,r)\hat{\phi}(y,r,z)\,dy - b(x,r)\hat{\phi}(x,r,z) \tag{1}$$

where $\hat{\phi}=\hat{\phi}(x,r,z)$ is the volume fraction density function, $p(x,y,r)$ is the fragment size distribution, $b(x,r)$ is the breakage frequency, $u(r)$ is mean axial velocity. x, y, r, z are the independent variables and x, y are the particle diameters, r is the radial and z is the axial coordinate.

The breakage frequency, $b(x,r)$, is considered to be made up of two functions; product of size dependent and r-dependent functions (Nere and Ramkrishna, 2005)

$$b(x,r) = b_1(x)b_2(r) \tag{2}$$

A homogenous breakage process describing the evolution of particle size distribution in time for a batch process can be described according to the expression (Ramkrishna, 2000)

$$\frac{\partial\phi(x,t)}{\partial t} = \int_x^\infty p(x,y)b_1(y)\phi(y,t)\,dy - b_1(x)\phi(x,t) \tag{3}$$

In this equation $\phi=\phi(x,t)$ is the volume fraction density function defined on the domain of particles of diameter x at time t. $p(x,y)$ which is known as breakage kernel or daughter size distribution, is the probability distribution of particles of size x resulting from the breakup of particles of size y and $b_1(x)$ is the breakage frequency (the breakage rate) at which particles of size x break per unit time. The integral part of Eq. (3) is the birth rate of particles and it describes the production of daughter size particles by breakage of parent size particles. The death rate of particles, which describes the average volume fraction of particles lost from the domain of particles of diameter x, is formulated as $b(x)\phi(x,t)$ in Eq. (3).

The relationship between the breakage equation for the cylindrical pipe system (Eq. (1)) and for the batch system (Eq. (3)) can be defined as (Kostoglou, 2006)

$$\hat{\phi}(x,r,z) = \phi\left(x, \frac{b_2(r)}{u_z(r)}z\right) \tag{4}$$

In the current work, the case of uniform velocity $u(r)=u$ and space independent fragment size distribution $p(x,y,r)=p(x,y)$ was examined in the cylindrical pipe. The solution of the well-known batch breakage equation was carried out by applying the discretization method.

3. Application of PBM to the breakage of whey protein precipitate particles

Alpha-lactalbumin enriched whey protein precipitates can be formed by the preferential reversible denaturation of the alpha-lactalbumin proteins in whey via pH and temperature adjustment (Pearce, 1983, 1987; Maubois and Ollivier, 1997). The resulting product can serve as a useful ingredient in the production of infant formula as well as nutritionally enhanced foods and sports performance drinks (De Wit, 1994). At an industrial scale, the precipitates may be separated from the supernatant by a process involving relatively high level of shear which incorporates operations such as pumping or centrifugal separation (Bell and Dunnill, 1982; Hoare et al., 1982; Bell et al., 1983). This study is based on work by Zumaeta et al. (2006) which attempted to mimic the high-shear environment of the separation process through the use of pipeline capillary flow of the precipitates where the flow intensity and the degree of breakage of the why protein precipitates were recorded. Zumaeta et al. (2006) developed a breakage model for whey protein precipitate particles by performing CFD simulations and comparing with experimental results for a whey protein precipitate dispersion flowing through a pipeline. In this work, breakage of whey protein precipitates passing through a pipeline with a constant length (Fig. 1) is examined at different velocities. A population balance is used here to model and predict such breakage processes. In their work, Zumaeta et al. (2006) investigated the effect of residence time of the particles within the pipe and the flow intensity on the breakage of whey protein precipitates particles.

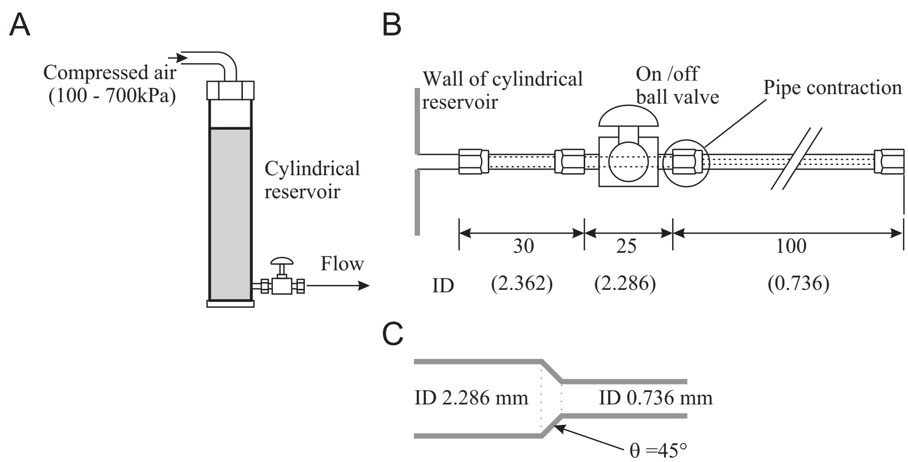

Fig. 1. Flow geometry: (A) metallic cylinder, (B) the 100 mm straight pipe, and (C) details of the smooth contraction (ID equals internal diameter; all dimensions in mm).

Table 1
Flow characteristics of whey protein precipitates propelled through 100 mm straight pipe (Zumaeta et al., 2006).

Mass flow rate ($g\,s^{-1}$)	Velocity ($m\,s^{-1}$)	Pressure (kPa)	Reynolds number (Re)
2.40	5.64	100	4100
4.44	10.46	300	7700
5.94	13.96	500	10 300
7.15	16.78	700	12 400

Table 2
The range of the values of coefficients c_1, c_2, c_3 and c_4.

Coefficient	Lower limit	Optimum	Upper limit
c_1	3460	3642	3824
c_2	16.27×10^{-4}	9.11×10^{-4}	5.06×10^{-4}
c_3	3833	4035	4237
c_4	0.292	0.343	0.392

3.1. Materials and methods

Alpha-lactalbumin rich whey protein precipitates were formed through temperature and pH adjustment via addition of HCl in a standard configuration agitation vessel. 1% *w/w* dispersions were formed under various agitation speeds. The resultant aggregate dispersion was aged for a 50 minute period in the agitation vessel before being propelled through a purpose built high-shear rig to mimic industrial processing conditions. The dispersion was recycled through the rig a number of times: 1, 2, 3, 4, 5, 10, 15, 20, 25, 30, 40, 50 and the particle size distribution of the samples were recorded. In the experiment, the dispersions were formed subject to impeller agitation at 550 rpm and were propelled through a 100 mm long straight pipe of diameter 0.736 mm at a number of different flow rates. Trials were carried out by applying compressed air driving pressures of 100, 300, 500 and 700 kPa which resulted in flow velocities of 5.64, 10.46, 13.96 and 16.78 $m\,s^{-1}$, respectively. The accompanying flow characteristics are shown in Table 1. A more detailed description of the materials and methods employed is presented in Zumaeta et al. (2006).

3.2. Breakage functions

The breakage functions which include the breakage frequency, $b(x,r)$, and the fragment size distribution, $p(x,y)$, are the core elements of PBEs. In this work, the breakage frequency which is based on the whey protein dispersions flowing through the turbulent pipe in a solid–liquid system, is derived from the experimental data of Zumaeta et al. (2006).

The functions $b_1(x)$ and $b_2(r)$ are taken to be a function of the cube root of particle volume, i.e. diameter, x, and a function of the turbulent energy dissipation rate, $\varepsilon(r)$. The function $b_1(x)$ has been reported to be a function of the particle diameter with the breakage coefficient of shear rate (Boadway, 1978; Pandya and Spielman, 1982; Peng and Williams, 1994). Moreover, Zumaeta et al. (2006) has shown that the breakage is an exponentially changing function of the shear rate. On this basis, when the experimental data, which are based on a number consecutive of recycles through the assembled rig (Fig. 1) at various velocities, are fitted the following expressions were found:

$$b_1(x) = c_1 e^{-c_2 G} x \tag{5a}$$

$$b_2(r) = c_3 e^{c_4 \varepsilon(r)} \tag{5b}$$

where $c_1 = 3642$, $c_2 = 9.11 \times 10^{-4}$, $c_3 = 4035$, $c_4 = 0.343$ and G is the average spatial shear rate in the vessel during aggregate formation and ageing which for the system and agitation speed employed (550 rpm in a 1.4 L standard configuration agitation vessel) can be represented by 700 s^{-1}. The level of significance was chosen as 0.05 for statistical analysis. That is, 95% significance interval is obtained for each coefficient. The coefficients c_1, c_2, c_3 and c_4 were calculated to obtain maximum likelihood between the model and experimental results. Regarding to 95% confidence interval, ranges which give satisfactory fits of the data for the values of these coefficients is given in Table 2.

Additionally, $\varepsilon(r)$ is taken to be the average energy dissipation rate during the process for the pipeline having internal diameter of 0.74 mm. These parameters are chosen due to their influence on particle strength and breakage as ascertained and described in previous work (Byrne and Fitzpatrick, 2002; Zumaeta et al., 2005, 2006). Correspondingly, Eq. (5a) and (5b) can be re-written as

$$b_1(x) = 1940x \tag{6a}$$

$$b_2(\varepsilon(r)) = 4035 e^{0.343\varepsilon(r)} \tag{6b}$$

On the other hand, two assumptions are made for the fragment size distribution: (i) binary breakage occurs, i.e. particles always break into two fragments as a result of a breakage event, (ii) the broken particles are uniformly distributed with respect to volume size. That is, broken particles have an equal probability of being between the smallest possible size (primary particle size) and their parent's size. Thus, the number density based fragment size distribution for volume size can be defined

$$p_n(\upsilon,\omega)=\frac{2}{\omega} \tag{7}$$

where ω and υ are respective particle volumes before and after a given breakage event and, the average number of particles resulting from a breakage of a particle of volume ω equals 2:

$$\int_0^{\omega}\frac{2}{\omega}\,d\upsilon=2 \tag{8}$$

Since the particle size under consideration is diameter and particles are assumed to be both spherical and of homogenous density, then volumes ω and υ can be characterized in the form of particle diameters y and x, respectively,

$$\omega=\frac{\pi}{6}y^3 \quad \text{and} \quad \upsilon=\frac{\pi}{6}x^3 \tag{9}$$

so, the fragment size distribution should be converted to be a function of diameter sizes by using the following relationship:

$$p_n(\upsilon,\omega)d\upsilon=p_{nd}(x,y)\,dx \tag{10}$$

which yields

$$p_{nd}(x,y)=\frac{6x^2}{y^3} \tag{11}$$

However, the fragment size distribution defined in Eq. (11) is still in number density based form. To convert to the volume based fragment size distribution a relationship between number density based fragment size distribution and volume density based fragment size distribution is required. This is defined

$$p_{nd}(x,y)=\frac{y^3}{x^3}p(x,y) \tag{12}$$

Thus the volume density based fragment size distribution with respect to diameter size has the following form:

$$p(x,y)=\frac{6x^5}{y^6} \tag{13}$$

Note that the function $p(x,y)$ should satisfy the mass conservation requirement such that all of the volume fractions formed sum to unity. It can be expressed as the following equation (McGrady and Ziff, 1986):

$$\int_0^{y}p(x,y)\,dx=1 \tag{14}$$

The solution to Eq. (3) in which Eqs. (6a) and (13) are inserted is obtained by discrete formulations which are discussed in the next section.

4. Discretized population balance for solution of breakage equation

The analytical solution of population balance equations of integro-differential form is not a trivial matter except for some simple examples. To overcome this difficulty, discrete methods can be employed to solve integro-differential type PBEs. There are a number of numerical techniques presented in the literature (Hill and Ng, 1995; Kumar and Ramkrishna, 1996; Vanni, 1999; Diemer and Olson, 2002; Marchisio et al., 2003). The underlying basis of these numerical techniques is to turn the main population balance of integro-partial differential equations into an ordinary differential equation by applying discretization as in the following type of equation:

$$\frac{d\phi_i}{dt}=\sum_{j=i+1}^{\infty}p_{ji}b_{1,j}\phi_j-b_{1,i}\phi_i \tag{15}$$

where ϕ_i is the volume fraction of particles in interval i containing particles larger than x_{i-1} and smaller than or equal to x_i. The time t is calculated according to Eq. (4). The discretized breakage frequency $b_{1,i}$ is the breakage frequency of i_{th} interval having representative diameter x_i has the form

$$b_{1,i}=1940x_i \tag{16}$$

The discretized volume based breakage function p_{ji} is the volume fraction of particles broken from interval j that go to interval i. Any broken particle in interval j cannot go into interval j, i.e.

$$\sum_{i=1}^{j-1}p_{ji}=1 \tag{17}$$

The restriction in Eq. (17) is frequently used for the discretized breakage equation in the literature. Consequently, the discretized fragment size distribution can be written by the following equation:

$$p_{ji}=\int_{x_{i-1}}^{x_i}p(x,x_{j-1})\,dx \tag{18}$$

4.1. Discretizing the size domain

Classification of a continuous size range into discrete intervals is the first step in constructing a discretized population balance equation. Two main approaches can be used to select the classification of the states.

- Uniform discretization: According to this approach, the interval of each state is constant. If the particle size is taken as its diameter, then $\check{r}=x_i-x_{i-1}$.

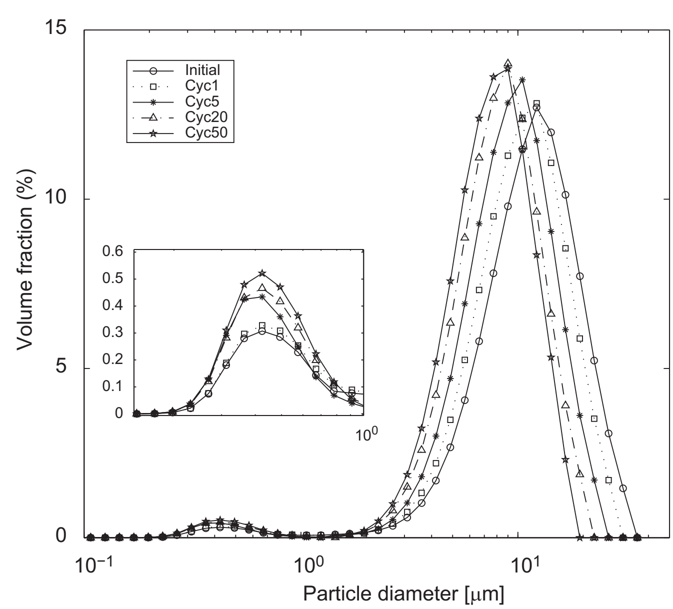

Fig. 2. Discrete raw data of volume fractions for whey protein precipitates subject to various numbers of cycles in shearing rig. The inset represents an enlargement of the submicron region.

- Geometric discretization: In this approach, the interval of the next state is directly proportional to its previous state. If the particle size is its diameter, then $\check{r} = x_i/x_{i-1}$,

where x_{i-1} and x_i are lower and upper limits of the interval i, respectively.

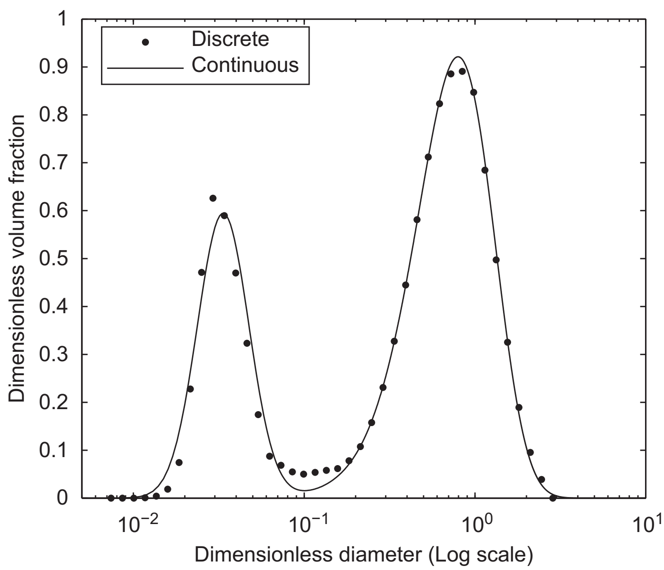

Fig. 3. Continuous and discrete density distribution of volume fractions for whey protein precipitates prior to shear rig exposure (initial conditions, dimensionless).

Although uniform discretization is numerically more stable it may lead a very large number of intervals. On the other hand, the number of intervals that result with geometric discretization can be much more reasonably dealt with and stable results may be obtained provided that the value of $\check{r}$ is well chosen.

The experimental data attained by Zumaeta et al. (2006) range from 0.085 to 38.4 μm with the number of intervals equal to 40. The particle diameter was discretized by using the geometric discretization with a consecutive diameter ratio $\check{r} = 1.65$. On account of this, the values for $b_{1,i}$ and p_{ji} were calculated according to Eqs. (16) and (18), respectively, by using discretized size ranges. On this basis all the information to solve Eq. (15) can be sourced to find the overall death rate for each interval. Once $d\phi_i/dt$ values are obtained, then the volume fraction, ϕ_i, of each interval can be easily calculated.

5. Results and discussion

The raw data generated by Zumaeta et al. (2006) include measurements of various number of cycles (1–5, 10, 15, 20, 25, 30, 40, 50) with flow velocities of 5.64, 10.46, 13.96 and 16.78 m s^{-1}. Fig. 2 displays an example of these raw data at initial, cycle 1, 5, 20 and 50 for a velocity 5.64 m s^{-1}.

For the purposes of this study all of the raw discrete data are normalized and nondimensionalized by: (i) dividing each volume fraction in particle size range by the total volume of particles (which is 100), and (ii) dividing each particle size by the initial mean particle size. Fig. 3 shows the resultant initial data which were taken from the measurements and displays the characteristic bimodal nature of the whey protein precipitate particle size distribution profile.

The discretized population balance method which was introduced in Section 4 is used to solve the breakage equation for whey protein

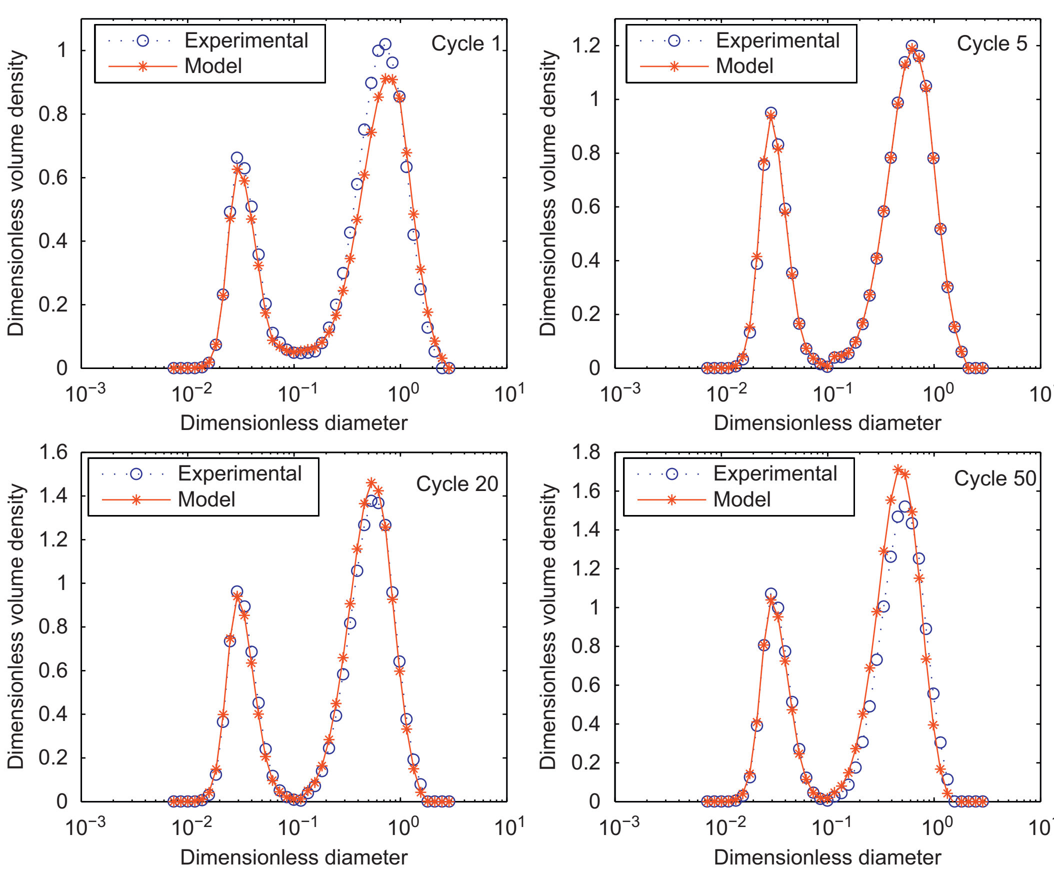

Fig. 4. Discrete volume fraction densities after cycles 1, 5, 20 and 50 at a velocity 5.64 m s^{-1}.

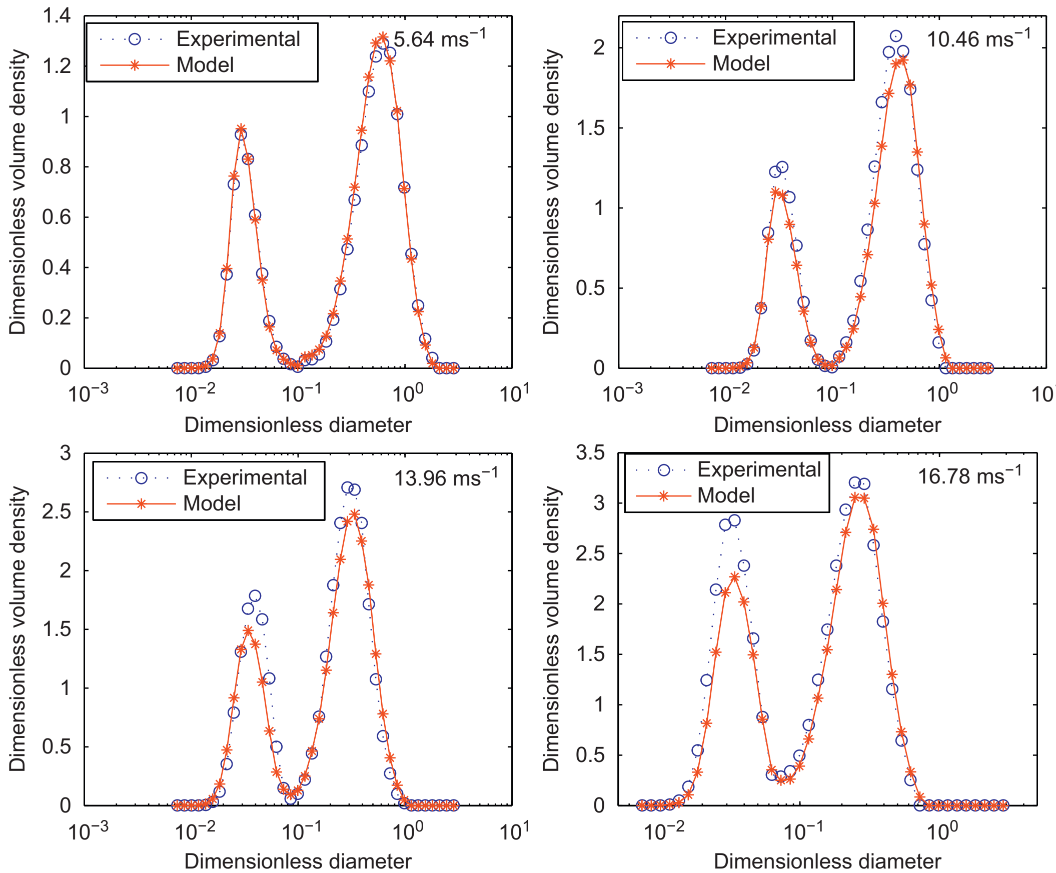

Fig. 5. Discrete volume fraction densities after cycle 10 at velocities 5.64, 10.46, 13.96 and 16.78 m s^{-1}.

particles. Accordingly, the results from the model are discrete in common with the experimental data and both have the same number of intervals thus making comparison easy.

Further calculations for the time intervals in Eq. (15) yield PSDs over a range of cycles and velocities. Comparisons of the model and the experimental results at velocity 5.64 m s^{-1} after various number of cycles are displayed in Fig. 4. It is clear that the model predictions show good agreement with the experimental results at various numbers of cycles. The breakage functions were also applied to the velocities of 10.46, 13.96 and 16.78 m s^{-1} which have the same initial condition as the velocity 5.64 m s^{-1}. The model predictions and the experimental results at various velocities after cycle 10 are presented in Fig. 5. These figures also show good correspondence between experimental results and the proposed model. When the model and experimental data are compared using a Kolmogorov–Smirnov test a correlation of at least 95% likelihood is displayed for all flow rates and cycles investigated.

In addition the mean particle sizes of each distribution given in Figs. 4 and 5 were used to demonstrate fractional error between predicted and experimental mean sizes for fines of dimensionless diameter less than 10^{-1}. Fractional errors were calculated by the ratio of the absolute difference between the expected mean size and the experimentally measured mean size to the experimentally measured mean size at a number of cycles

$$\textit{Fractional error} = \frac{|\textit{Expected mean size} - \textit{Experimental mean size}|}{\textit{Experimental mean size}} \tag{19}$$

The resultant values are given in Tables 3 and 4.

Table 3
Fractional errors between predicted and experimental mean sizes for fines of dimensionless diameter less than 10^{-1} at a number of cycles for the velocity 5.64 m s^{-1}.

Cycle	Expected mean size (dimensionless)	Experimental mean size (dimensionless)	Fractional error (dimensionless)
1	0.0433	0.0436	0.007
5	0.0368	0.0368	0.000
20	0.0386	0.0379	0.018
50	0.0381	0.0383	0.005

Table 4
Fractional errors between predicted and experimental mean sizes for fines of dimensionless diameter less than 10^{-1} at cycle 10 for velocities 5.64, 10.46, 13.96 and 16.78 m s^{-1}.

Velocity (m s^{-1})	Expected mean size (dimensionless)	Experimental mean size (dimensionless)	Fractional error (dimensionless)
5.64	0.0369	0.0373	0.011
10.46	0.0393	0.0395	0.005
13.96	0.0449	0.0441	0.018
16.78	0.0456	0.0466	0.021

On the other hand, it is observed that, the mean particle size is decreasing while volume fraction is increasing with increasing high-shear capillary exposure time (number of recycles). Additionally the volume fractions of particles associated with smaller (submicron) sizes increase with increasing capillary exposure time as would be expected with ongoing breakage event.

Moreover, the model may be employed as a useful predictive tool for various PSD profiles such as for different velocities and for

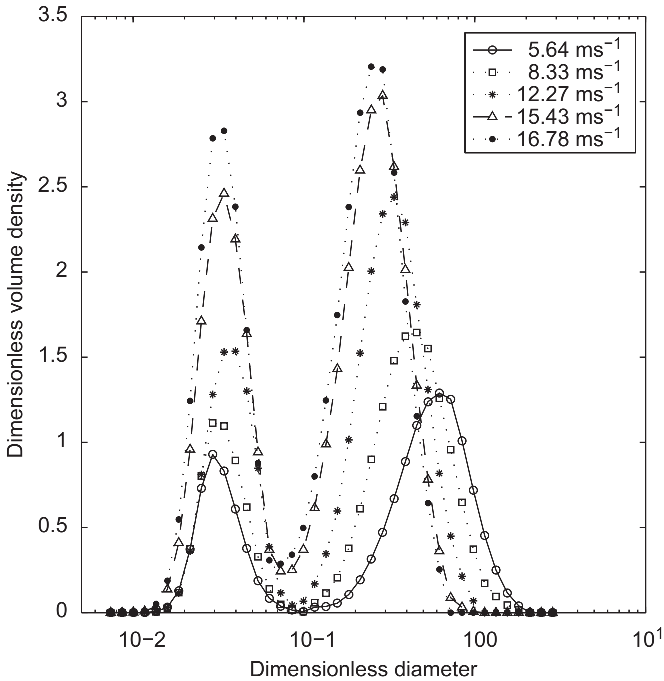

Fig. 6. Comparison of discrete volume fractions after cycle 10 with different velocities.

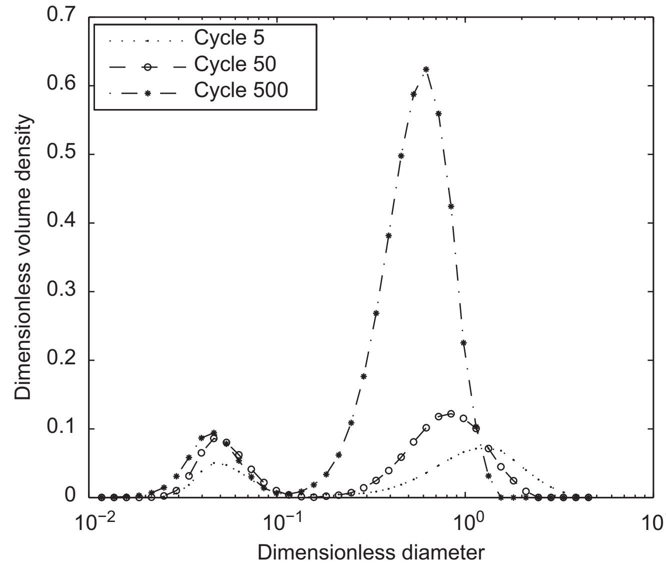

Fig. 7. Volume density fractions for cycle 5, 50 and 500 at a velocity $5.64\,\mathrm{m\,s^{-1}}$.

different corresponding flow rates. Fig. 6 shows PSDs predicted by the model subjected to velocities of 5.64, 8.33, 12.27, 15.43 and $16.78\,\mathrm{m\,s^{-1}}$ after 10 cycles. In this figure, predicted PSDs corresponding to velocities of 8.33, 12.27 and $15.43\,\mathrm{m\,s^{-1}}$ are also displayed and even though these velocities are not applied experimentally. Nevertheless they show distributions consistent with adjacent experimental data in terms of the characteristic bimodal distribution of the particles. In general it can be shown that the mean size of particles decreases with increasing applied velocity. The increment in velocity has a positive effect on breakage frequency by means of the flow velocity.

Additionally, the model was also applied to cycle 500 at the velocity $5.64\,\mathrm{m\,s^{-1}}$ as an example to demonstrate particle size distribution after a large number of cycles. Fig. 7 shows PSD profiles of cycle 5, 50 and 500. According to this figure, it can be interpreted that the breakage of bigger particles occurs more than the breakage of fine particles. The bimodal distribution of the particles is still preserved even after cycle 500.

6. Conclusion

The determination of the particle size distribution for breakage processes has an important role in many engineering applications. The evolution of size distribution of particles can vary from one system to another and to follow this size evolution is important to design and optimize the process. Population balances can be used to describe such distributions. The structure of these equations are rigorous and since they exist as partial integral differential equations their solution may not always be feasible. However numerical methods for the solution of a breakage equation may realize results just as significant as those of analytical ones.

In this paper, the discrete method for a breakage equation which is one of the basic numerical methods for the solution of population balance equations, is applied to the experimental results of Zumaeta et al. (2006). The selection of an appropriate breakage frequency and an appropriate fragment size distribution play a very important role in the model. In fact, they are central to the model and the selection of these functions is the most important part of the modelling procedure. Breakage frequency is a function of both historic shear conditions during aggregation and flow velocities through the high-shear capillary rig as proposed by Zumaeta et al. (2006). In the case of this study the selected breakage frequency and the fragment size distribution representing the breakage of whey protein precipitates justify the model. The model successfully tracks the experimental PSDs over the applied velocity range and is also used to predict PSD for other nontested flow rates.

Notation

$b(x,r)$	the breakage frequency, $\mathrm{s^{-1}}$
G	the shear rate, $\mathrm{s^{-1}}$
p	the volume density based fragment size distribution with respect to diameter size, dimensionless
p_n	the number density based fragment size distribution with respect to volume size, dimensionless
p_{ji}	the discrete fragment size distribution, dimensionless
p_{nd}	the number density based fragment size distribution with respect to diameter size, dimensionless
P	the cumulative fragment size distribution, dimensionless
r	the radial coordinate
$\check{r}$	a consecutive diameter ratio
t	the time, s
u	the mean axial velocity, $\mathrm{m\,s^{-1}}$
x	the particle diameter, dimensionless
y	the parent particle diameter, dimensionless
z	the axial coordinate

Greek letters

ε	the turbulent eddy dissipation rate, $\mathrm{m^2\,s^{-3}}$
υ	the particle volume, dimensionless
ϕ	the volume fraction density function, dimensionless
ω	the parent particle volume, dimensionless

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Appendix A. Finding a continuous density function

The continuous density function can be found by applying some probability distribution such as normal distribution, beta distribution, lognormal distribution and gamma distribution. Clearly experimental results in Fig. 3 displays a characteristic bimodal distribution. As a result of statistical analysis of these data it was found that the lognormal distribution is appropriate to define the volume fraction distribution of whey protein precipitates which have diameters smaller then or equal to 1.15 μm and the gamma distribution is appropriate to define the volume fraction distribution of whey protein precipitates which have diameters bigger then 1.15 μm. Accordingly, the initial case of this bimodal distribution can be defined by the following density function:

$$\phi_0(x) = q\frac{1}{\sigma x\sqrt{2\pi}}\exp\left(-\frac{(\ln(x)-\mu)^2}{2\sigma^2}\right) + (1-q)x^{\alpha-1}\frac{\beta^\alpha\exp(-\beta x)}{\Gamma(\alpha)} \tag{A.1}$$

where σ and μ are scale and location parameters of lognormal distribution, respectively, and α and β are scale and rate parameters of gamma distribution, respectively. q is weight of the first component of bimodal distribution and has a value $0 < q < 1$ which is calculated as 0.0176 from the initial distribution of whey protein precipitates having diameters smaller then or equal to 1.15 μm (i.e. for lognormal part of bimodal distribution). The corresponding initial continuous density distribution of discrete volume fraction distribution is shown in Fig. 3. Note that the main property of probability distributions is that the area under the continuous curve equals 1. The concerned system is represented by 40 intervals and sum of volume fractions of all intervals at any stage of the process equals 1 by reason of the mass conservation. This can be formulated as

$$\sum_{i=1}^{40}\phi_i = 1 \tag{A.2}$$

The first component of bimodal distribution covers intervals from 1 to 17. Thus, the value of q for lognormal distribution can be calculated according to

$$q = \sum_{i=1}^{17}\phi_i \tag{A.3}$$

Note that q values for different number of cycles and at various velocities should be updated according to the bimodal distribution obtained.

In addition, if a valid probability density function for volume distribution is found it is easy to obtain the probability density function for number distribution provided that particles are spherical and of homogenous density. When $\phi(x,t)$ is volume fraction density function, a number density function $n(x,t)$ can be calculated using the following equation:

$$n(x,t) = \frac{\left(\phi(x,t)/\frac{\pi}{6}x^3\right)}{\int_0^\infty\left(\phi(x,t)/\frac{\pi}{6}x^3\right)dx} \tag{A.4}$$

The denominator in Eq. (A.4) is essential to normalize the area under $n(x,t)$ to 1 as it is a fundamental property of a PDF.

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